

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR  
MID-Spring Semester 2024-25

Date of Exam: .09.24  
Subject Number: MA60225  
Subject Name: Financial Mathematics

Session(FN/AN)  
Max. Marks: 30

Duration: 2 hrs  
Department: Mathematics  
No. of students: 126

- Instructions:** (i) Use of calculator and Statistical tables is allowed.  
(ii) All answers of numerical questions must be in at least two decimal places.  
(iii) All the notations are standard. If any data/statement is missing, identify it on your answer script.  
(iv) Answer All questions. Marks are indicated at the end of each part of question.  
(v) All parts of a question Must Be answered at One Place.

1. For this question (Question 1 only) write only answers on the first page of your answer script in the given tabular form. Detail working may be carried out on other pages. [1×6]

| Q. No. 1 | a | b | c | d | e | f | g | h | i | j |
|----------|---|---|---|---|---|---|---|---|---|---|
| Answer   |   |   |   |   |   |   |   |   |   |   |

- (a) A stock is presently selling at a price of Rs 75 per share. After one time period, its selling price will (in present value dollars) be either Rs 200 or Rs 50. An option to purchase  $y$  units of the stock at time 1 for the (present value) price  $120y$  costs  $cy$ . What should  $c$  be in order for there to be no sure win?
- (b) Consider random walk  $X_0 := 0$ ,  $X_n = Y_1 + \dots + Y_n$ ,  $n \geq 1$ ;  $Y_j$ 's are i.i.d. random variables with  $E|Y_1| < \infty$ . The stochastic process  $\{X_n\}_{n \geq 0}$  is a Martingale if  $E(Y_1) =$
- (c) Let  $B(t)$  be a Brownian motion starting at 0 with drift  $\mu = 0$  and variance  $\sigma^2 = 1$ . Find the probability that  $B(t)$  reaches level  $-2$  before reaching 1.
- (d) Let  $\{W_t, t \geq 0\}$  be a Wiener process. Then, find the value of  $\text{Var}(W_4) \times \text{Cov}(W_1, W_4)$ .
- (e) Compute the stochastic differential  $d\cos(W_t)$ .
- (f) Let  $W_t$  be a Wiener process and  $T_a$  denote the time it takes this process to hit  $a$ . Then find  $P(T_a < \infty)$ .
2. Write short note on following: [2×3]
- (a) Call options.
- (b) Put options.
- (c) No arbitrage principle.

3. (a) Explain dynamic programming technique in Markov decision processes. Explain the optimality equation in dynamic programming, derive it using the total probability formula. [3]

- (b) Consider the random walk that in each  $\Delta t$  time unit either goes up or down the amount  $\sqrt{\Delta t}$  with respective probabilities  $p$  and  $1 - p$ , where  $p = \frac{1}{2}(1 + \mu\sqrt{\Delta t})$ . Argue that as  $\Delta t \rightarrow 0$  the resulting limiting process is a Brownian motion process with drift rate  $\mu$ . [3]

4. (a) Express the solution  $X_t$  of the following stochastic differential equation as a stochastic integral with respect to a Wiener process  $W_t$ .

$$dX_t = -X_t dt + dW_t, \quad X_0 = x_0,$$

where  $x_0$  is a constant. Whether it is an OU process? Explain. Also find  $E(X_t)$  and  $Var(X_t)$ . [3]

- (b) Let

$$X_t := e^{W_t - \frac{t}{2}}, \quad t \geq 0,$$

where  $\{W_t\}_{t \geq 0}$  be a Wiener process.

- (i) Whether  $X_t$  is an martingale (MG)? Explain.

- (ii) Apply Itô's formula to compute the stochastic differential  $dX_t$ . [3]

5. (a) Derive and explain single period Binomial market. [3]

- (b) Consider a single-period binomial financial market with  $\Omega = \{\omega_1, \omega_2\}$ , with the current (time  $t = 0$ ) asset price  $S_0 = 5$  and  $B_0 = 1$ , and the terminal price (time  $t = 1$ ) stock prices  $S_1(\omega_1) = 6$  and  $S_1(\omega_2) = 40/9$ . Assume interest rate  $r = 1/9$ .

- (i) Show that the market is arbitrage-free.

- (ii) Consider a contingent claim  $X$  with  $X(\omega_1) = 6$  and  $X(\omega_2) = 1$ . Find the arbitrage-free value of the claim at time  $t = 0$ .

- (iii) Construct a replicating portfolio for the claim  $X$  and verify that its values at times  $t = 0$  and  $t = 1$  coincides with the claim price you found in part (ii) and the claim value  $X$ , respectively. [3]

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